Experiment 1.4: Muon Physics

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Report

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Abstract

The mean lifetime of cosmic-ray muons stopped in a plastic scintillator was measured by recording the time interval between the muon entry pulse and the subsequent decay positron pulse with an FPGA-based timer. Over approximately 43 hours of acquisition, the lifetime was determined to be $\tau_{\text{obs}} = 2.09 \pm 0.07$ μs, consistent with a population-weighted average of the free-space positive muon lifetime and the shorter negative muon lifetime in carbon arising from nuclear capture. The Fermi weak interaction constant was extracted as $G_F/(\hbar c)^3 = (1.140 \pm 0.019) \times 10^{-5}$ GeV^{-2}, in agreement with the accepted value within two standard uncertainties. The muon charge ratio could not be determined with statistical significance, as the lifetime uncertainty exceeded the difference between τ^+ and τ^-.

1 Introduction

High-energy particles from the universe, called cosmic rays, enter and interact with Earth's atmosphere to create a shower of high-energy particles, including charged pions ($\pi^{\pm}$). Some of these cosmic rays originate from the sun, but most from the wider universe from supernovae explosions, black holes and other cosmic events and objects [1]. Cosmic rays are doubly beneficial to astronomy and particle physics; providing a buffet of interesting particles to study whose energies and properties encode information about their origin. These high-energy particles also prove mischievous, as they can cause 'single-event upsets' by interacting with electronics and causing a bit-flip; impacting the on-board computers of airplanes and spacecraft; or casting electoral votes for the 2003 Belgian federal election [2].

Charged pions decay via the weak interaction with a lifetime of 26.02 ns [3] into muons ($\mu^{\pm}$) and (anti-)neutrinos (ν_μ, $\bar{\nu}_\mu$) as

$$\pi^+ \to \mu^+ + \nu_\mu$$
$$\pi^- \to \mu^- + \bar{\nu}_\mu$$

The μ^+ and μ^- particles have the same approximate mass of 105.66 MeVc^{-2} [4] and mean kinetic energies of 4 GeV [4]. They decay further into electrons ($e^{\pm}$) and (anti-)neutrinos ($\nu_{\mu,e}$, $\bar{\nu}_{\mu,e}$) via the weak interaction with a free-space mean lifetime of $\tau = 2.19703 \pm 0.00004$ μs [5] as

$$\mu^+ \to e^+ + \nu_e + \bar{\nu}_\mu$$
$$\mu^- \to e^- + \bar{\nu}_e + \nu_\mu$$

Due to the effect of time-dilation, the lifetime of muons created by pion decay is comparable to the time taken to reach the surface of the earth, hence muon decay events can be observed on the surface, providing an opportunity to determine the lifetime of the muons. Beyond confirming the Standard Model of particle physics, the muon lifetime is of interest because it directly determines the Fermi coupling constant G_F that characterises the strength of the weak interaction.

Additionally, determination of the ratio of positive to negative muons ratio can provide insight into the relative cross-sections of cosmic ray interactions. It is expected that more positive muons are created because most cosmic rays, which is mostly protons and nuclei, and the nuclei they interact with are positively charged [6], hence the product particles would be mostly positively charged due to conservation laws. Additionally, negative muons of the same energy are more likely to decay before reaching the detector given their ability to couple with nuclei however the effects of this in air are expected to be small given the low density.

2 Theory

2.1 The muon decay time distributions

Muon decays into electrons follow a Poisson distribution such that the probability of k decays in time t is given by

$$P(k,t) = \frac{(\lambda t)^k e^{-\lambda t}}{k!} \tag{2.1}$$

where $\lambda = 1/\tau$ is a constant decay rate, the inverse of the mean lifetime τ. The probability of observing a decay at time Δ after entering a detector at some $t = t_0$ is given by the decay time distribution

$$D(\Delta) = \lambda e^{-\lambda \Delta} \tag{2.2}$$

such that the function representing the histogram of decay events is given by

$$C = Ae^{-\Delta/\tau} + B \tag{2.3}$$

where A is a normalisation constant at B is a constant offset.

2.2 Interaction of muons with matter

Nucleus	Mean lifetime (μs)	Capture rate (s^{-1})
^{1}H	2.194903 ± 0.000066	420 ± 20
^{12}C	2.02 ± 0.02	$(36 \pm 4) \times 10^3$

Table 1: Modified table from [7] of the properties of μ^- with respect to the nuclei present in the scintillator. The literature values for the mean lifetime, and by extension the capture rate for ^{12}C vary between source. The value taken is from Sens [8], motivated by this paper also containing experimental value for the disappearance rate.

Due to their charge, negative muons can be captured by nuclei and enter bound states. The weak decay rate of muons into electrons is different in bound states than in free space and depends on the nuclei the muon is bound to. The detector used in this experiment is a scintillator filled with Polyvinyltoluene, a synthetic polymer with empirical formula C_9H_{11}. It can be seen by comparison of the capture rates for C, H in table 1 that the effects of carbon nuclei will dominate this behaviour. The negative muon lifetime is expected to be nearly equal to the lifetime in carbon of $\tau_C = 2.02 \pm 0.02 \mu s$ [8].

Captured muons have small atomic radii due to their large mass [4] and hence interact with protons via the weak force as

$$\mu^- + p \rightarrow n + \nu_\mu$$

where it is noted that the kinetic energy of the neutron is significantly less than the energy of an electron created by spontaneous neutron decay, given neutrons are ≈ 1800 times more massive than electrons [9]. Under the assumptions that the muon interacts independently with all protons, the nucleus can be modelled as a point and the interaction is proportional to muon density at the position of the nucleus, a dependence probability on nuclear charge, Z, is constructed to be Z^4. Muons that undergo this proton interaction are not detected, since high-energy electrons are not produced. The nuclear branching fraction, which is defined as the proportion of negative muons in carbon nuclei bound states that interact with protons as above, as opposed to decaying into electrons, is found by Sens as $8.9 \pm 0.2\%$ of incident muons [8].

2.2.1 Muon charge ratio

The average observed decay rate, $\langle\lambda\rangle$, for positive and negative muons is

$$\langle\lambda\rangle = \frac{N^+\lambda^+ + x\ N^-\lambda^-}{N^+ + x\ N^-} = \frac{\rho\lambda^+ + x\ \lambda^-}{\rho + x} \tag{2.4}$$

where $\lambda^\pm$ is the decay rate for $\mu^\pm$, $N^\pm$ is the number of muons incident on the scintillator per unit time, $\rho = N^+/N^-$ is the charge ratio and x is the disappearance rate. The charge ratio can be deduced as

$$\rho = -\frac{\tau^+}{\tau^-}\left(\frac{x\ (\tau^- - \tau_{obs})}{\tau^+ - \tau_{obs}}\right) \tag{2.5}$$

from a single measurement of τ.

2.3 Fermi's Coupling Constant

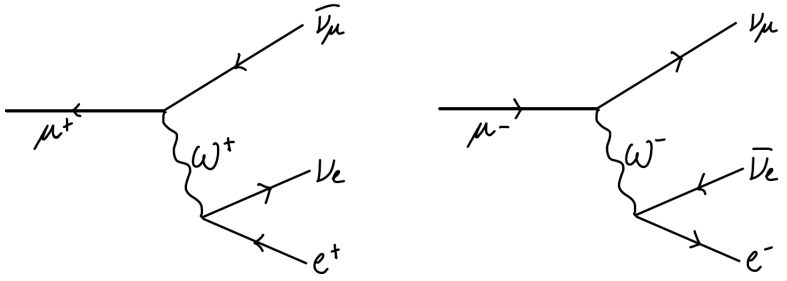


Figure 1: Feynman diagram of the decay of positive and negative muons into electrons and anti-neutrinos through the weak force.

The first-order Feynman diagrams for the decay of muons is depicted in figure 1. As per the Standard Model, the muon lifetime is related to the Fermi weak interaction constant, G_F as

$$\frac{1}{\tau} = \frac{1}{\hbar}\frac{G_F^2}{(\hbar c)^6}\frac{(m_\mu c^2)^5}{192\pi^3} \tag{2.6}$$

where m_μ is the rest mass of a muon, $105.66 MeV c^{-2}$ [4].

3 Experimental Methods

3.1 Experimental Setup

3.1.1 Detector

The detector is a cylindrical plastic scintillator with $d = 0.15$ m, $h = 0.125$ m. The incident muons at the surface have an expected flux from cosmic rays of 180 $\mathrm{m}^{-2}\mathrm{s}^{-1}$ [4]. The rate of muons expected to enter the detector from above is $n = 3.18$ Hz, however the total rate of muons is expected to be slightly higher, given that muons do not approach perfectly perpendicular to the earth's surface.

Incident muons in the scintillator rapidly deposit kinetic energy to the scintillator resulting in a light pulse. Muons that decay in the detector produce highly energetic electrons, as muons are ≈ 207 times more massive than electrons. These electrons also generate a light pulse in the detector as they lose energy such that two pulses are expected for each decay event, and the time between them is measured as the value Δ.

3.1.2 Electronics

The detector system block diagram is pictured in figure 2. A Digital Storage Oscilloscope (DSO) and a multimeter, not pictured, are used to investigate and calibrate the electronic component behaviour.

The FPGA acts as a precise timer. When a muon enters the detector setup, a pulse is detected and the timer starts. If a decay event happens in the scintillator, such that a second pulse is created within a pre-defined period of time, the FPGA stops the timer and sends this information to the PC. A specialised software is used to store the timestamped outputs of the FPGA timer.

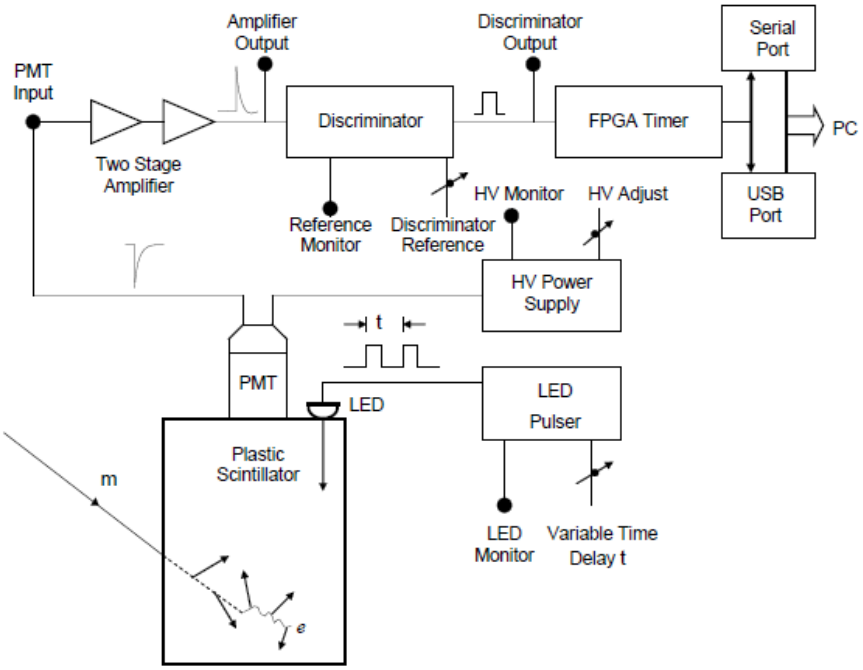


Figure 2: Figure taken from [4]. Light pulses in the scintillator are amplified by the photomultiplier (PMT). The level of amplification is adjusted through the high voltage adjust (HVA). The output signal from the PMT is amplified by a two stage amplifier. The amplifier output is processed by a discriminator with a variable discriminator level. Signals above this level produce a square pulse output that is detected by the Field-Programmable Gate Array (FPGA) that relays information to the PC.

3.2 Electronics Calibrations

3.2.1 PMT and amplifier characterisation

The PMT high-voltage adjust was set to 634.00 ± 0.10 mV and its output observed on the DSO in averaging mode ($N = 512$). Single-photon events produced negative pulses of amplitude 459 ± 4 mV and width 66 ± 3 ns, with pulse-to-pulse amplitude variation consistent with the stochastic nature of the dynode multiplication chain.

The amplifier gain was characterised by injecting a 100 mV peak-to-peak sine wave from the function generator into the PMT input and sweeping the frequency from 100 Hz to 2.5 MHz in logarithmic steps. Both input and amplifier output were 50 Ω terminated. Additionally, the amplifier output has a 1050Ω resistor on the output which divides the measured voltage by a factor of 21. This was corrected in post-processing.

3.2.2 Discriminator characterisation

With a 100 kHz, 100 mV peak-to-peak sine wave at the PMT input, the discriminator threshold was swept across its full range. The output width was found to increase as the threshold was lowered, and the discriminator saturated high for thresholds above the sine-wave peak. With the scintillator reconnected, muon-candidate events appeared as a characteristic short primary pulse followed by a longer secondary pulse; the threshold was then raised until non-conforming noise events were suppressed.

3.2.3 FPGA timing validation

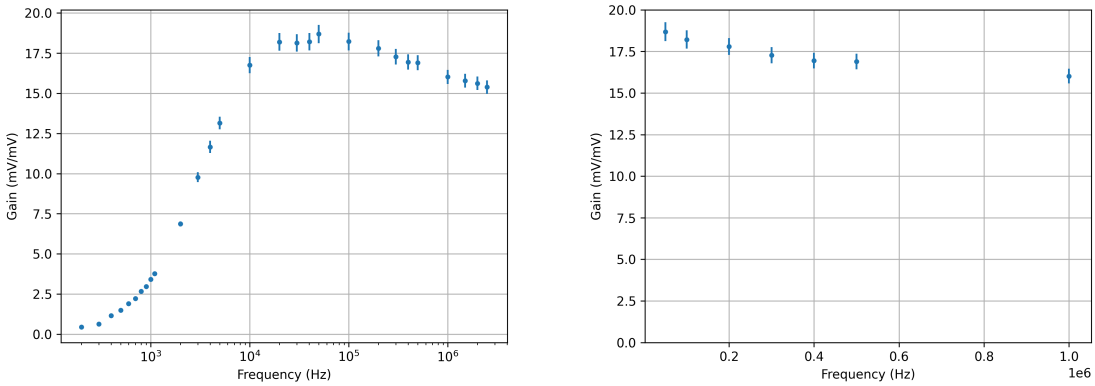
The pulser was used to inject pairs of pulses of known separation, and ΔT as reported by the FPGA-based acquisition software was compared against the DSO "Period" measurement across the range 1–20 μs. The two measurements agreed within their combined uncertainties at every point, validating the FPGA timing for the lifetime measurement.

3.3 Data Acquisition

The discriminator threshold was set to $198.50 \pm 0.10\,\text{mV}$, chosen to give a muon detection rate of approximately $3\ \text{s}^{-1}$ while suppressing noise triggers. The pulser was disabled, the PMT output connected to the electronics box, and the acquisition software configured with a 20-bin histogram. ΔT values were logged continuously over approximately 43h.

4 Results

4.1 Gain functionality



(a) (b)

Figure 3: Amplifier gain (mV/mV) against frequency (Hz) with error bars against a. a logarithmic x-axis and b. a linear x-axis windowed to relevant frequencies for the experiment of $(5 \times 10^4) - 10^6$ Hz. The trend suggests unreliable amplifier function at lower frequencies and comparatively consistent gain for frequencies relevant to the experiment.

A plot of gain against frequency for the amplifier is produced in figure 3. The gain of the amplifier is observed to increase significantly with frequency until it plateaus in the $10^4 - 10^5$ Hz range, after which it shows a slight decline. In this experiment, pulse separations from 1μs to 20μs are investigated, corresponding to frequencies of $(5 \times 10^4) - 10^6$ Hz. The gain decreases in a linear fashion from 18.7 ± 0.6 to 16.0 ± 0.4 across this range.

4.2 Muon Lifetime

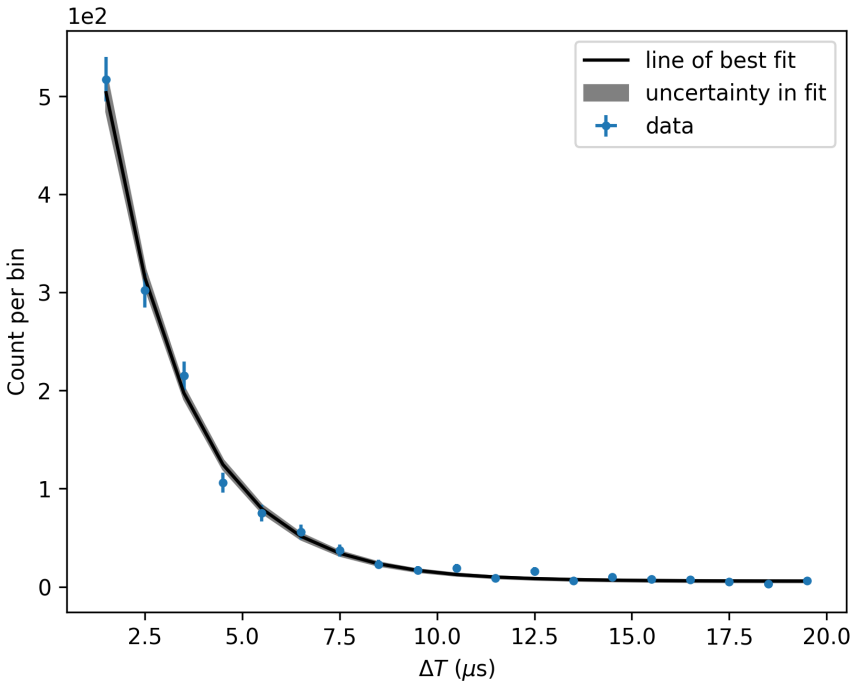


Figure 4: A histogram of the number of muon decay events recorded within each time interval ΔT is plotted with a bin size of 1 μs with error bars from Poisson statistics. An exponential line of best fit in black is a strong fit to the data. The reduced χ^2 for the fit is 1.5.

The recorded muon event data is cleaned of spurious event peaks, with significantly higher count numbers in a time interval than the perceived average over the data collection period. An increased count was observed for the $0-1$ μs bin compared to the general trend of the data, likely to correspond to unfiltered electronic noise. This range was excluded from the data as an outlier. A histogram of the clean data was produced with a bin width of 1 μs and fit according to equation 2.3 with weighted uncertainties to find the mean muon lifetime to be 2.09 ± 0.07 μs. Alternative bin widths were explored, it was determined that higher resolution than 1 μs was not statistically significant enough and lower resolutions, particularly 2μs produced χ^2 values less than 1, indicating insufficient data points for the determination of the fit parameters.

The muon charge ratio was calculated as per equation 2.5 with the literature values $\tau_+ = 2.1969803 \pm 0.0000022$ μs [5], $\tau_- = \tau_C = 2.02 \pm 0.02$ μs [8] and the experimental value $\tau_{obs} = 2.09 \pm 0.07$ μs. The outcome of this calculation was $\rho = \frac{N^+}{N^-} = 0.6 \pm 1.1$, with standard propagated uncertainties.

Fermi's Coupling Constant was determined from the experimental data as per equation 2.6 to be $G_F/(\hbar c)^3 = (1.140 \pm 0.019) \times 10^{-5}$ GeV^{-2}.

5 Discussion

The lifetime of muons in the scintillator was successfully determined as $2.09 \pm 0.07\mu$s by measuring the timing of muon decay events. This value was expected, and is between the free-space decay time and the decay time in carbon, however the uncertainty is comparable to the difference in these values, rendering a calculation of the muon charge ratio statistically insignificant. This value is calculated as $\frac{N^+}{N^-} = 0.6 \pm 1.1$, which agrees with the value found by Khachatryan et al. [6] of 1.2766 ± 0.0032 by a direct measurement, using a magnetic field to separate positive and negative incident muons. However, the uncertainty of the measured value is higher than the actual measured value, so this is not considered to be a strong indicator of agreement.

The precision of the measured lifetime is fundamentally limited by the total acquisition time of approximately 43 hours, which determines the number of decay events collected and hence the count statistics in each histogram bin; this uncertainty

propagates strongly into the derived value of ρ. It would be necessary to record muon data for a more extended period of time, or with a significantly larger scintillator to capture a larger flux, in order to find a muon lifetime value that is statistically significant enough to determine the muon charge ratio. It would also be beneficial to decrease electronic noise, such as with an electromagnetic shield around the detector system. An increase in the number of events detected would also allow for better timing resolution, as smaller bin margins could be applied to the data. It is observed that the detector system has a much higher resolution than is used in the experiment, in the order of $20ns$. The detector electronics can hence fulfil this higher resolution if sufficient counts were collected. The limitation of the slow frequency of muon decay events from cosmic rays, and the uncertainty introduced by negative muon interactions is why the precise measurements of muon decay do not come from measuring cosmic rays, but rather from experiments with positive muon beams [10].

The muon decay lifetime is used to find a value for Fermi's weak interaction constant of $G_F/(\hbar c)^3 = (1.140 \pm 0.019) \times 10^{-5}$ GeV^{-2}, two standard uncertainties above the literature value of $G_F/(\hbar c)^3 = 1.1663785 \times 10^{-5}$ GeV^{-2} [10]. Muon decay is generally an ideal scenario to study the weak interaction, given it is the only force present in the decay of free-space muons. However, it is known that some negative muons in this experiment decay from bound states for which the interaction strength is not described purely by equation 2.6, thus the value found for the Fermi weak interaction constant is affected. A statistically significant measurement of the muon charge ratio could provide insight into the significance of this interference.

The muon decay data was filtered to remove spurious events. Spurious events could be caused by either a cosmic ray shower, causing a momentary increase in the rate of muons from the atmosphere, or from electronic noise. Numerous spurious event peaks were observed at the start of data acquisition, and two were observed at times closer to the end time of the experiment. It is to be noted that the data acquisition started at around 3pm on Tuesday April 21st and was concluded around 10:30am on Thursday April 23rd. The lab sessions, during which numerous different experiments are run, go from 10am–4pm on Tuesdays and Thursdays. Most of the spurious events detected line up with the time that the lab was expected to be open, suggesting electronic noise from other apparatus as the likely cause rather than cosmic ray showers, and justifying their exclusion from the dataset. This noise hypothesis is further supported by the elevated count in the 0-1 $mathrm\mu s$ bin, which is inconsistent with the dead-time hypothesis: if the FPGA were unable to record short-interval events, a deficit rather than an excess would be expected. The elevated count is therefore also attributed to noise not fully rejected by the discriminator threshold.

The investigation into the amplifier functionality reveals limited variation in the gain of the amplifier system across relevant frequencies of (5×10^4)–10^6 Hz. This section of the experiment was repeated due to an error in the terminator setup. The natural impedance in the BNC connector cables of $Z = 50\ \Omega$ requires a terminator to prevent reflection of the incoming signal. The amplifier input to the DSO initially did not have a 50 Ω terminator connected so the signal was being reflected back along the connecting BNC cable. The trend line for this incorrect setup mimicked that in figure 3, with the exception of an increase for high frequency inputs > 1 MHz. This increase for high frequencies is caused by the reflected wave constructively interfering with the incoming wave, amplifying the signal.

To improve upon this experiment, it would be beneficial alter the scintillator in order to remove the decay contributions from bound μ^- decay. Kratochwil et al. have successfully produced a scintillator from thallium chloride (TlCl) doped with beryllium (Be) or iodine (I) [11]. Assuming the dopant level is low, the expected nuclear branching fraction, d, the proportion of captured μ^- that decay via proton interaction as opposed to spontaneous decay, weighted by the capture rate for each element is estimated for this material as

$$d = \frac{a_{\mathrm{Tl}}\, d_{\mathrm{Tl}} + a_{\mathrm{Cl}}\, d_{\mathrm{Cl}}}{a_{\mathrm{Tl}} + a_{\mathrm{Cl}}} \tag{5.1}$$

where a is the capture rate. The nuclear branching fraction and capture rate for Tl and Cl are in table 2. The resulting branching fraction for nuclear capture is 0.94 ± 0.07, indicating that approximately 94% of stopped μ^- are captured before decaying. The measured lifetime in such a scintillator would therefore be dominated by free-space muon decay, allowing a cleaner extraction of τ and, by comparison with the present measurement, the muon charge ratio.

Nucleus	Capture rate (s^{-1})	Nuclear branching fraction
Tl	$(12.90 \pm 0.75) \times 10^6$	0.96 ± 0.08
Cl	$(1.39 \pm 0.09) \times 10^6$	0.76 ± 0.06

Table 2: Capture rates [7] and disappearance rates [8] for the nuclei in TlCl. Note that the disappearance rate for Cl is from Cl(Na).

6 Conclusion

The mean lifetime of muons in the plastic scintillator was measured as $\tau_{\text{obs}} = 2.09 \pm 0.07$ μs, consistent with the expected value lying between the free-space positive muon lifetime and the lifetime of negative muons bound in carbon. This result was used to derive the Fermi weak interaction constant as $G_F/(\hbar c)^3 = (1.140 \pm 0.019) \times 10^{-5}$ GeV^{-2}, agreeing with the literature value within two standard uncertainties. The muon charge ratio could not be determined with statistical significance, as the uncertainty on τ_{obs} exceeded the difference between τ^+ and τ^-. A longer acquisition period, a larger scintillator volume, or a high-Z stopping material such as TlCl doped with Be,I to suppress the bound μ^- contribution would each enable a meaningful determination of ρ in future iterations of this experiment.

References

[1] P. A. Mason and P. L. Biermann, "Astrophysical and cosmological constraints on life," in *Habitability of the Universe Before Earth*, ser. Astrobiology Exploring Life on Earth and Beyond, Academic Press, 2018, pp. 89–126, ISBN: 978-0-12-811940-2. DOI: https://doi.org/10.1016/B978-0-12-811940-2.00005-8. [Online]. Available: https://www.sciencedirect.com/science/article/pii/B9780128119402000058.

[2] J. Caulfield. "When the sun casts its vote," Next Nature Network, Accessed: May 22, 2026. [Online]. Available: https://nextnature.org/en/magazine/story/2017/sun-influences-elections.

[3] A. J. Greenberg et al., "Charged-pion lifetime and a limit on a fundamental length," *Phys. Rev. Lett.*, vol. 23, pp. 1267–1270, 21 Nov. 1969. DOI: 10.1103/PhysRevLett.23.1267. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevLett.23.1267.

[4] *Lab script 1.4: Muon physics*, PHS3000 Physics Laboratory, Monash University, 2026.

[5] Particle Data Group, "July 2010 particle physics booklet," *Journal of Physics G: Nuclear and Particle Physics*, vol. 37, p. 075 021, 2010. DOI: 10.1088/0954-3899/37/7A/075021.

[6] V. Khachatryan et al., "Measurement of the charge ratio of atmospheric muons with the cms detector," *Physics Letters B*, vol. 692, no. 2, pp. 83–104, 2010, ISSN: 0370-2693. DOI: https://doi.org/10.1016/j.physletb.2010.07.033.

[7] T. Suzuki, D. F. Measday, and J. P. Roalsvig, "Total nuclear capture rates for negative muons," *Phys. Rev. C*, vol. 35, pp. 2212–2224, 6 Jun. 1987. DOI: 10.1103/PhysRevC.35.2212. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevC.35.2212.

[8] J. C. Sens, "Capture of negative muons by nuclei," *Phys. Rev.*, vol. 113, pp. 679–687, 2 Jan. 1959. DOI: 10.1103/PhysRev.113.679. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRev.113.679.

[9] P. J. Mohr, D. B. Newell, B. N. Taylor, and E. Tiesinga, "Codata recommended values of the fundamental physical constants: 2022," *Rev. Mod. Phys.*, vol. 97, p. 025 002, 2 Apr. 2025. DOI: 10.1103/RevModPhys.97.025002. [Online]. Available: https://link.aps.org/doi/10.1103/RevModPhys.97.025002.

[10] D. M. Webber et al., "Measurement of the positive muon lifetime and determination of the fermi constant to part-per-million precision," *Phys. Rev. Lett.*, vol. 106, p. 041 803, 4 Jan. 2011. DOI: 10.1103/PhysRevLett.106.041803. [Online]. Available: https://link.aps.org/doi/10.1103/PhysRevLett.106.041803.

[11] N. Kratochwil et al., "TlCl:Be,I: A high sensitivity scintillation and Cherenkov radiator for TOF-PET," *IEEE Transactions on Radiation and Plasma Medical Sciences*, vol. 9, no. 3, pp. 296–303, Mar. 2025, Epub 2024 Oct 31. PMID: 40045986; PMCID: PMC11882110. DOI: 10.1109/TRPMS.2024.3487359.

7 Appendix

7.1 AI Use Statement

In addition to the AI usage for preparation of the logbook, I used Claude/ClaudeCode in the following ways in preparing the work presented here:

1. First draft of concise method from logbook notes.
2. Re-wording the conclusion.
3. Writing the abstract.

4. Error checking.